

COM3067 – ALGORITHMS

MIDTERM EXAM QUESTIONS – 15.11.2025

1. (30p) Given two arrays of lengths n , a conceptual two-dimensional $n \times n$ array is formed whose rows and columns are sorted in increasing order. An integer x is also provided. The following brute-force algorithm is employed:

```
BruteForce( $A, x$ )  
  for  $i \leftarrow 0$  to  $n - 1$   
    for  $j \leftarrow 0$  to  $n - 1$   
      if  $A[i, j] = x$  then  
        return true  
  return false
```

(a) Briefly explain what this algorithm does.

(b) Design a decrease-and-conquer algorithm that performs the same task in $O(n)$ running time.

2. (30p) There are n pancakes of all different sizes that are stacked on top of each other. You are allowed to slip a flipper under one of the pancakes and flip over the whole stack above the flipper. The purpose is to arrange pancakes according to their size with the biggest at the bottom. Design an algorithm that runs in $O(n^2)$ time.

3. (25p) A graph has been explored by a DFS algorithm; and the discovery and finishing times of its vertices are given as A(1, 14), B(3, 4), C(10, 13), D(2, 5), E(6, 9), F(7, 8), and G(11, 12). Using only this information, construct an adjacency matrix for a possible graph.

4. (15p) Solve the following recurrence relation using Master Theorem.

$$T(n) = \begin{cases} 1, & n = 0, \\ 5T\left(\frac{n}{2}\right) + n^2, & n > 0. \end{cases}$$